

IBM – AI

MAJOR PROJECT

Project 2: House Price Prediction System Objective:

Predict house prices based on various features like area and location.

Key Concepts:

- 1 Regression Models
- 2 Feature Engineering
- 3 Model Evaluation Metrics (R^2 , MAE)

Steps:

- 1 Load dataset using Pandas
- 2 Handle missing values
- 3 Encode categorical variables
- 4 Train regression model
- 5 Evaluate and predict results

Output: Predicted house price and comparison graph

DataSet:

1. Here is the reference to access the – house_data file



CODE:**1. Here is the code for performing House Prediction System**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer

#
=====

# Step 1: Load YOUR dataset
#
=====

script_dir = os.path.dirname(os.path.abspath(__file__))
df = pd.read_csv(os.path.join(script_dir, 'house_data.csv'))

print("--- Column Names ---")
print(df.columns.tolist())
print("\n--- First 5 Rows ---")
print(df.head())
```

```

#
=====

# Step 2: Handle missing values
#
=====

print("\n--- Missing Values BEFORE ---")
print(df.isnull().sum())

# Fill missing numbers with median, missing text with most frequent value
for col in df.select_dtypes(include=[np.number]).columns:
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(include=['str']).columns:
    df[col] = df[col].fillna(df[col].mode()[0])

print("\n--- Missing Values AFTER ---")
print(df.isnull().sum())

#
=====

# Step 3: Encode categorical variables
#
=====

# CHANGE these to match YOUR column names
numerical_features = ['area', 'bedrooms', 'age']
categorical_features = ['location']
target_column = 'price'

```

```
X = df[numerical_features + categorical_features]
```

```
y = df[target_column]
```

```
preprocessor = ColumnTransformer(
```

```
    transformers=[
```

```
        ('num', 'passthrough', numerical_features),
```

```
        ('cat', OneHotEncoder(drop='first'), categorical_features)
```

```
    ])
```

```
X_encoded = preprocessor.fit_transform(X)
```

```
# Split data
```

```
X_train, X_test, y_train, y_test = train_test_split(X_encoded, y,
    test_size=0.2, random_state=42)
```

```
#
```

```
=====
```

```
# Step 4: Train regression model
```

```
#
```

```
=====
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
#
```

```
=====
```

```
# Step 5: Evaluate and predict
```

```

#
=====

y_pred = model.predict(X_test)

r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)

print(f"\n--- R2 Score: {r2:.4f} ---")
print(f"--- Mean Absolute Error: ${mae:,.2f} ---")

results = pd.DataFrame({
    'Actual Price': y_test.values[:10],
    'Predicted Price': np.round(y_pred[:10], 2)
})

print("\n--- Sample Results ---")
print(results)

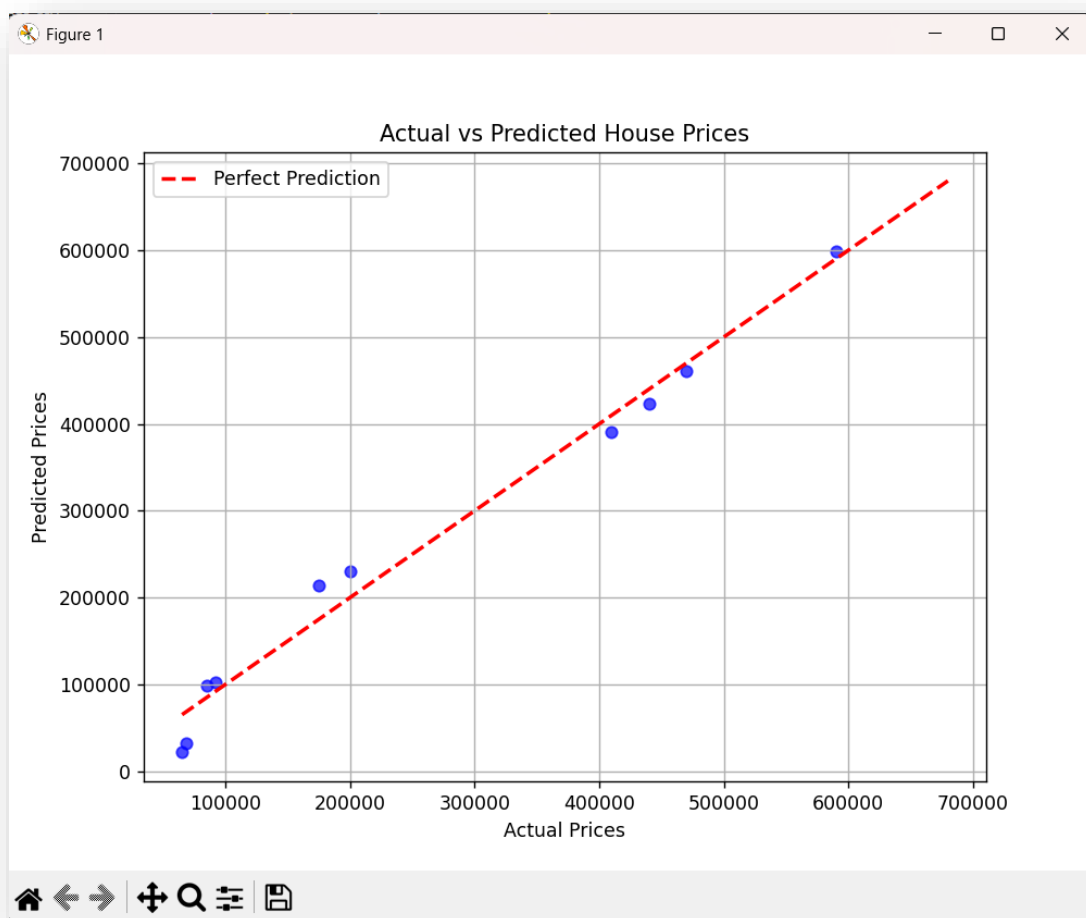
# Graph: Actual vs Predicted
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.7, color='blue')
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual vs Predicted House Prices')
plt.legend()
plt.grid(True)

```

```
plt.show()
```

OUTPUT:

1. Here is the output of the comparison graph



2. Here is the output of the terminal code

a) Missing Values Check

b) Model Evaluation Metrics

c) Sample Predicted Results

```
PS C:\Users\Mitun S\OneDrive\Desktop\IBM> & "C:\Users\Mitun S\AppData\Local\Python\pythoncore-3.14-64\python.exe" "c:/Users/Mitun S/OneDrive/Desktop/IBM/Major Project/major.py"
--- Column Names ---
['area', 'bedrooms', 'location', 'age', 'price']

--- First 5 Rows ---
   area  bedrooms  location  age  price
0  1500         3    Urban    5  450000
1   800         2    Rural   20  120000
2  2200         4  Suburban    2  520000
3  1200         2    Urban   10  350000
4  3000         5  Suburban    1  680000

--- Missing Values BEFORE ---
area      0
bedrooms  0
location  0
age       0
price     0
dtype: int64

--- Missing Values AFTER ---
area      0
bedrooms  0
location  0
age       0
price     0
dtype: int64

--- R² Score: 0.9810 ---
--- Mean Absolute Error: $22,589.08 ---

--- Sample Results ---
   Actual Price  Predicted Price
0      470000      461489.91
1      68000      32095.61
2     410000     390100.88
3      85000      98953.16
4     200000     230462.38
5     590000     598898.08
6     440000     423062.47
7      92000     101788.65
8     175000     213226.70
9      65000      21689.31
PS C:\Users\Mitun S\OneDrive\Desktop\IBM>
```

TOOLS AND TECHNOLOGY:

Language: Python 3

- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- IDE: Visual Studio Code

CONCLUSION:

The Linear Regression model successfully predicts house prices based on area, bedrooms, location, and age.

- The R^2 score indicates the model explains a significant portion of the variance in house prices.
- The scatter plot shows predicted values closely follow the actual prices, validating the model's performance.